

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular End Semester Examination – Summer 2022

Course: B. Tech.

Branch : Civil Engineering

Semester : IV

Subject Code & Name: BTCVC404 Water Resources Engineering

Max Marks: 60

Date: 24/08/2022

Duration: 3.45 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

- A) Explain the different methods of distribution of water. **COI Understand 6**
- B) After how many days will you supply water to soil in order to ensure sufficient irrigation of the given crop, if
- (i) Field capacity of the soil = 28%
 - (ii) Permanent wilting point = 13%
 - (iii) Dry density of soil = 1.3 gm/c.c.
 - (iv) Effective depth of root zone = 70 cm
 - (v) Daily consumptive use of water for the given crop = 12 mm
- Assume any other data not given.
- C) The gross command area for a distributary is 6000 hectares, 80% of which is culturable irrigable. The intensity of irrigation for Rabi season is 50% and that for Kharif season is 25%. If the average duty at the head of the distributary is 2000 hectares/cumec for Rabi season and 900 hectares/cumec for Kharif season, find out the discharge required at the head of the distributary from average demand considerations. **COI Apply 6**

Q.2 Solve Any Two of the following.

- A) What are the different Zones of storage/ control levels in a reservoir? Explain with the help of a diagram. **CO2 Understand 6**
- B) Analyse the following failures in Gravity dam:-
- a) By overturning (or rotation) about the toe
 - b) By crushing (or compression)
- C) A proposed reservoir has capacity of 500 ha-m. The catchment area is 125 km², and the annual stream flow averages 12 cm of runoff. If the annual sediment production is 0.03 ha.m/km², what is the probable life of the reservoir before its capacity is reduced by 10% of its initial capacity by sedimentation? The relationship between trap efficiency η (%)

C/I	0.01	0.02	0.04	0.06	0.08	0.1	0.2	0.3	0.5	0.7
η (%)	43	60	74	80	84	87	93	95	96	97

Q. 3 Solve Any Two of the following.

- A) Explain the components of earthen dam and their functions with the help of a diagram. **CO2 Understand 6**
- B) Write a short note on following failures in earthen dam:-
- a. Hydraulic Failure
 - b. Seepage Failure
 - c. Structural Failure
- CO3 Understand 6**

- C) What are the assumptions and limitations regarding Kennedy's silt theory?

CO3
Understand

6

Q.4 Solve Any Two of the following.

- A) A catchment has 6 raingauge stations. In a year, the annual rainfall recorded by the gauges are as follows:

Station	A	B	C	D	E	F
Rainfall (cm)	82.6	102.9	180.3	110.3	98.8	136.7

CO3
Apply

6

For a 10% error in the estimation of mean rainfall, calculate the optimum numbers of stations in the catchment.

- B) The ordinates of 3hr UH of a catchment are given below

Time (hr)	0	3	6	9	12	15	18	21
3 hr UHO (m ³ /s)	0	10	20	16	12	8	4	0

CO3
Apply

6

Derive flood hydrograph at the catchment outlet due to a storm given below. Assume Φ index is 3 mm/hr and constant base flow 10 m³/s.

Time (hr) for start of storm	0	3	6	9
Accumulated rainfall (cm)	0	3.9	4.7	7.6

- C) Explain the following methods to analyze rainfall record data with the help of diagram:
a. Mass Curve of rainfall
b. Hyetograph

CO3
Understand

6

Q. 5 Solve Any Two of the following.

- A) Explain groundwater movement using Darcy's law.
B) Explain Bligh's Creep Theory and its limitations.
C) What are the causes and ill-effects of water logging?

CO3
Understand

6

CO3
Understand

6

Understand

6

*** End ***